

SemEval-2024 Task 8: Multidomain, Multimodel and Multilingual Machine-Generated Text Detection

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Abstract

We present the results and the main findings of SemEval-2024 Task 8: Multigenerator, Multidomain, and Multilingual Machine-Generated Text Detection. The task featured three subtasks. Subtask A is a binary classification task determining whether a text is written by a human or generated by a machine. This subtask has two tracks: a monolingual track focused solely on English texts and a multilingual track. Subtask B is to detect the exact source of a text, discerning whether it is written by a human or generated by a specific LLM. Subtask C aims to identify the changing point within a text, at which the authorship transitions from human to machine. The task attracted a large number of participants: subtask A monolingual (126), subtask A multilingual (59), subtask B (70), and subtask C (30). In this paper, we present the task, analyze the results, and discuss the system submissions and the methods they used. For all subtasks, the best systems used LLMs.

1 Introduction

The proliferation of Large Language Models (LLMs) has led to a significant increase in the volume of machine-generated text (MGT) across a wide range of domains. This rise has sparked concerns regarding the potential for misuse in fields such as journalism, education, academia, etc (Uchendu et al., 2023; Crothers et al., 2023). Moreover, it poses challenges to maintaining information integrity and ensuring accurate information dissemination. As such, the ability to accurately distinguish between human-written content and machine-generated content has become paramount for identifying potential misuse (Jawahar et al., 2020; Stiff and Johansson, 2022; Macko et al., 2023).

In response to these challenges, we are introducing a shared task that focuses on the detection of machine-generated text across multiple generators,

domains, and languages. We are providing large-scale evaluation datasets for **three subtasks** with the primary goals of fostering extensive research in MGT detection, advancing the development of automated systems for detecting MGT, and reducing instances of misuse:

Subtask A: Human vs. Machine Classification.

The goal of this subtask is to accurately classify a text as either produced by a human or generated by a machine. This is the basic, but one of the most common use-cases of MGT detection systems for preventing the misuse of LLMs. This task is divided into two tracks: (i) The *monolingual track*, which focuses solely on English texts; and (ii) The *multilingual track*, which involves texts in a variety of languages, thereby expanding the diversity and complexity beyond existing benchmarks.

Subtask B: Multi-Way Generator Detection.

This task aims to pinpoint the exact source of a text, i.e., determine whether it originated from a human or a specific LLM (GPT-3, GPT-3.5, GPT-4, Cohere, DALL-E, or BLOOMz). Determining a particular LLM that potentially generated the given text is important from several perspectives: it can help to narrow down the set of LLMs for more sensitive white-box detection techniques or in cases where the generated material is harmful, misleading, or illegal, it might be useful for addressing ethical concerns and legal obligations.

Subtask C: Changing Point Detection.

The goal of this subtask is to precisely identify the exact boundary (changing point) within a text at which the authorship transitions from a human to machine happens. The texts begin with human-written content, which at some point is automatically continued by LLMs (GPT and LLaMA series). The percentage of the human-written section varies from 0% to 50%. This task takes into account the fact

that in many malignant use-cases of LLMs, the part of the text might be written by a human and a part might be generated by a machine. It is hard to classify a text as machine-generated if a big chunk is actually human-written. This is a way to obscure the usage of LLM, and the formulation of Subtask C addresses this challenge.

The task attracted a large number of participants: 126 teams for the Subtask A monolingual track, 59 teams for the Subtask A multilingual track, 70 teams for Subtask B, and 30 teams for Subtask C, with a total of 54 participating teams having submitted a system description paper for all subtasks.

Next, we introduce the MGT detection techniques considered in this shared task in §2; §3 describes the corpus and the evaluation metrics; §4 details the organization of the task; §5 provides an overview of the participating systems; and §6 discusses the evaluation results.

2 Background

Detecting machine-generated text is primarily formulated as a binary classification task (Zellers et al., 2019; Gehrmann et al., 2019a; Solaiman et al., 2019; Ippolito et al., 2019), naively distinguishing between human-written and machine-generated text. In general, there are two main approaches: the supervised methods (Wang et al., 2024a,b; Uchendu et al., 2021; Zellers et al., 2019; Zhong et al., 2020; Liu et al., 2022) and the unsupervised ones, such as zero-shot methods (Solaiman et al., 2019; Ippolito et al., 2019; Mitchell et al., 2023; Su et al., 2023; Hans et al., 2024). While supervised approaches yield relatively better results, they are susceptible to overfitting (Mitchell et al., 2023; Su et al., 2023). Meanwhile, unsupervised methods may require unrealistic white-box access to the generator. In the following, we provide background information on each subtask, respectively.

Subtask A: Mono-lingual and Multi-lingual Binary Classification Given the prevalence of the binary classification task, various benchmarks assess model performance in both mono-lingual and multi-lingual settings. HC3 (Guo et al., 2023) compares ChatGPT-generated text with human-written text in English and Chinese, utilizing logistic regression models trained on GLTR Test-2 features (Gehrmann et al., 2019a) and RoBERTa (Liu et al., 2019)-based classifiers for detection. Benchmark results by Wang et al. (2024b) include

evaluations of several supervised detectors, such as RoBERTa (Liu et al., 2019), XLM-R (Conneau et al., 2019), logistic regression classifier with GLTR features (Gehrmann et al., 2019b), and stylistic features (e.g., stylometry (Li et al., 2014), NELA (Horne et al., 2019) features). Macko et al. (2023) create a similar resource called MULTI-TuDE for 11 languages in the news domain and conduct an extensive evaluation of various baselines. Our effort extends the previous works by providing evaluation setup for multiple domains, multiple languages, and for state-of-the-art LLMs, including ChatGPT and GPT-4.

Subtask B: Multi-Way Generator Detection

Multi-way generator detection, attributing texts not just to their machine-generated nature but also to specific generators, resembles authorship attribution. Munir et al. (2021) find that texts from language models (LMs) have distinguishable features for source attribution. Uchendu et al. (2020) addresses three authorship attribution problems: (1) determining if two texts share the same origin, (2) discerning whether a text is machine or human-generated, and (3) identifying the language model responsible for text generation. Approaches like GPT-who by Venkatraman et al. (2023) employ UID-based features to capture unique signatures of each language model and human author, while Rivera Soto et al. (2024) leverages representations of writing styles.

Subtask C: Change Point Detection Change point detection, which is closely tied to authorship obfuscation (Macko et al., 2024), extends beyond binary/multi-class classification to an adversarial co-authorship setting involving both humans and machines (Dugan et al., 2023). Machine-generated text detection methods are vulnerable to authorship obfuscation attacks such as paraphrasing (Crothers et al., 2022; Krishna et al., 2023; Shi et al., 2023; Koike et al., 2023), back-translation, and change point detection. Related to Subtask C, (Gao et al., 2024) introduces a dataset with mixed machine and human-written texts using operations such as polish, complete (Xie et al., 2023), rewrite (Shu et al., 2023), humanize (adding natural noise (Wang et al., 2021)), and adapt (Gero et al., 2022). Kumarage et al. (2023) uses stylometric signals to quantify changes in tweets and detect when AI starts generating tweets. Different to our task, they focus on human-to-AI author changes within a given Twitter

Split	Source	davinci-003	ChatGPT	Cohere	Dolly-v2	BLOOMz	GPT-4	Machine	Human
Train	Wikipedia	3,000	2,995	2,336	2,702	–	–	11,033	14,497
	WikiHow	3,000	3,000	3,000	3,000	–	–	12,000	15,499
	Reddit	3,000	3,000	3,000	3,000	–	–	12,000	15,500
	arXiv	2,999	3,000	3,000	3,000	–	–	11,999	15,498
	PeerRead	2,344	2,344	2,342	2,344	–	–	9,374	2,357
Dev	Wikipedia	–	–	–	–	500	–	500	500
	WikiHow	–	–	–	–	500	–	500	500
	Reddit	–	–	–	–	500	–	500	500
	arXiv	–	–	–	–	500	–	500	500
	PeerRead	–	–	–	–	500	–	500	500
Test	Outfox	3,000	3,000	3,000	3,000	3,000	3,000	18,000	16,272

Table 1: **Subtasks A: Monolingual Binary Classification.** Data statistics over Train/Dev/Test splits

Split	Language	davinci-003	ChatGPT	LLaMA2	Jais	Other	Machine	Human
Train	English	11,999	11,995	–	–	35,036	59,030	62,994
	Chinese	2,964	2,970	–	–	–	5,934	6,000
	Urdu	–	2,899	–	–	–	2,899	3,000
	Bulgarian	3,000	3,000	–	–	–	6,000	6,000
	Indonesian	–	3,000	–	–	–	3,000	3,000
Dev	Russian	500	500	–	–	–	1,000	1,000
	Arabic	–	500	–	–	–	500	500
	German	–	500	–	–	–	500	500
Test	English	3,000	3,000	–	–	9,000	15,000	13,200
	Arabic	–	1,000	–	100	–	1,100	1,000
	German	–	3,000	–	–	–	3,000	3,000
	Italian	–	–	3,000	–	–	3,000	3,000

Table 2: **Subtasks A: Multilingual Binary Classification.** Data statistics over Train/Dev/Test splits (Others generators are Cohere, Dolly-v2 and BLOOMz)

timeline.

3 Dataset and Metrics

In this section, we describe the datasets and evaluation metrics for all subtask tracks, including the size, domains, generators, and language distribution across training, development, and test splits.

3.1 Subtask A: Monolingual Track

Data: Table 1 presents statistics across generators, domains, and splits. The training set encompasses domains such as Wikipedia, WikiHow, Reddit, arXiv, and PeerRead, comprising a total of 56,400 machine-generated and 63,351 human-written texts. BLOOMz is utilized as an unseen generator in the development set, which contains 2,500 machine-generated and 2,500 human-written texts. For the test set, OUTFOX is introduced as the surprising domain, and GPT-4 serves as the surprising generator, with a dataset of 18,000 machine-generated and 16,272 human-written texts.

Metrics: Accuracy is used to evaluate detectors.

3.2 Subtask A: Multilingual Track

Data: Table 2 presents the dataset statistics. The training set encompasses texts in English, Chinese, Urdu, Bulgarian, and Indonesian, totaling 76,863 machine-generated and 80,994 human-written texts. The development set includes Arabic (sourced

Split	Source	davinci-003	ChatGPT	Cohere	Dolly-v2	BLOOMz	Human
Train	Wikipedia	3,000	2,995	2,336	2,702	2,999	3,000
	WikiHow	3,000	3,000	3,000	3,000	3,000	2,995
	Reddit	3,000	3,000	3,000	3,000	2,999	3,000
	arXiv	2,999	3,000	3,000	3,000	3,000	2,998
Dev	PeerRead	500	500	500	500	500	500
Test	Outfox	3,000	3,000	3,000	3,000	3,000	3,000

Table 3: **Subtasks B: Multi-Way Generator Detection.** Data statistics over Train/Dev/Test splits

Domain	Generator	Train	Dev	Test	Total
PeerRead	ChatGPT	3,649 (232)	505 (23)	1,522 (89)	5,676 (344)
	LLaMA-2-7B*	3,649 (5)	505 (0)	1,035 (1)	5,189 (6)
	LLaMA-2-7B	3,649 (227)	505 (24)	1,522 (67)	5,676 (318)
	LLaMA-2-13B	3,649 (192)	505 (24)	1,522 (84)	5,676 (300)
	LLaMA-2-70B	3,649 (240)	505 (21)	1,522 (88)	5,676 (349)
OUTFOX	GPT-4	–	–	1,000 (10)	1,000 (10)
	LLaMA2-7B	–	–	1,000 (8)	1,000 (8)
	LLaMA2-13B	–	–	1,000 (5)	1,000 (5)
	LLaMA2-70B	–	–	1,000 (19)	1,000 (19)
Total	all	18,245	2,525	11,123	31,893

Table 4: **Subtask C: Change Point Detection.** We use generators GPT and LLaMA-2 series over domains of academic paper review (PeerRead) and student essay (OUTFOX). The number in “()” is the number of examples purely generated by LLMs, i.e., human and machine boundary index=0. LLaMA-2-7B* and LLaMA-2-7B used different prompts. Bold data is used in shared task training, development, and test.

from Wikipedia), Russian, and German (sourced from Wikipedia), each contributing 2,000 texts from both machine-generated and human-written sources. In the test set, Italian is introduced as the unexpected language, with *OUTFOX* and *News* serving as new domains for English, Arabic, and German texts. This set comprises 22,100 machine-generated and 20,200 human-written texts.

Metrics: Accuracy is used to evaluate detectors.

3.3 Subtask B

Data: In Table 3, we incorporate texts from five generators (davinci-003, ChatGPT, Cohere, Dolly-v2, and BLOOMz) alongside human-written texts. The development set features texts from the PeerRead domain, while the test set introduces OUTFOX (specifically, student essays) as the unexpected domain.

Metrics: Accuracy is used to evaluate detectors.

3.4 Subtask C

Data: The training and development sets for subtask C are PeerRead ChatGPT generations, with **5,349** and **505** examples respectively (first row of Table 4), and the test set is the combination of the *test column* of Table 4, totaling 11,123 examples.

Metrics: The Mean Absolute Error (MAE) is used to evaluate the performance of the boundary detection model. It measures the average absolute difference between the predicted position index and the actual changing point.

4 Task Organization

The shared task was run in two phases:

Development Phase. Only training and development data were provided to the participants, with no gold labels available for the development set. Participants competed against each other to achieve the best performance on the development set. A live leaderboard on CodaLab was made available to track all submissions. Teams could make an unlimited number of submissions, and the best score for each team, regardless of the submission time, was displayed in real time on CodaLab.

Test Phase. The test set was released, containing two additional languages—German and Italian for Subtask A Multilingual Track, generator GPT-4 for the Monolingual Track, and a new domain (student essays) for Subtask B. For Subtask C, both new domains and generators were introduced (GPT-4 and LLaMA-2 series based on PeerRead and OUTFOX), which were not disclosed to the participants beforehand (referred to as surprise languages, domains, and generators).

Participants were given approximately three weeks to prepare their predictions. They could submit multiple runs, but they wouldn’t receive feedback on their performance. Only the latest submission from each team was considered official and used for the final team ranking.

In total, 125 teams submitted results for Subtask A Monolingual, 62 for Subtask A Multilingual, 70 for Subtask B, and 30 for Subtask C. Additionally, 54 teams submitted system description papers.

After the competition concluded, we released the gold labels for both the development and test sets. Furthermore, we kept the submission system open for the test dataset for post-shared task evaluations and to monitor the state of the art across the different subtasks.

5 Participating Systems

In this section, we first summarize common features for all teams based on the information they provided in the Google Docs. Then, we delve into

Team Name	Ranking	small PLM	LLM	GPT	fine-tuning	zero-shot	few-shot (k=?)	Data augmentation	External Data
Genaios	1		✓						
USTC-BUPT	2	✓			✓				
petkaz	12	✓			✓				
HU	17		✓		✓			✓	
TrustAI	20	✓			✓				
L3i++	25		✓		✓				
art-nat-HHU	26	✓			✓				
Unibuc - NLP	28	✓			✓			✓	
NewbieML	30	✓							
QUST	31		✓		✓			✓	
NootNoot	39	✓			✓				
Mast Kalandar	40		✓		✓				
I2C-Huelva	41		✓		✓				
Werkzeug	45	✓			✓				
NCL-UoR	50		✓		✓				
Sharif-MGTD	51		✓		✓	✓			
Collectivized Semantics	62	✓			✓				
SINAI	61		✓		✓				
MasonTigers	71	✓	✓		✓	✓			
DUTh	73		✓		✓				
surbhi	74		✓		✓				
KInIT	77		✓		✓	✓			
RUG-D	100		✓		✓				✓
RUG-5	101	✓	✓		✓				
RUG-3	114		✓	✓	✓				
Mashee	115		✓				2		
RUG-1	117	✓							

Table 5: **Subtask A monolingual** participants methods overview. *small PLM*: Pre-trained Language Model is used, *LLM*: LLM is used, *GPT* indicates if any GPT models are used, *fine-tuning*: applying fine-tuned models, *zero-shot* and *few-shot (k=?)* that zero or more examples are used as demonstrations in in-context learning based on LLMs, *Data augmentation* and *External Data* refers to that augmented data or other external data have been used.

the methods employed by the top 3 teams, accompanied by brief descriptions of the approaches utilized by the other top 10 teams.

The approaches of all teams are presented in Appendix A.

5.1 Monolingual Human vs Machine

Table 5 provides a high-level overview of the methodologies employed by the top-ranking systems in Subtask A monolingual. Most systems utilized either a Pretrained Language Model (PLM) or a Large Language Model (LLM), with the majority of participants fine-tuning their models. Usage of GPT, external data, and few-shot methods was observed in only one team each.

Team Genaios_{STA_mono:1} (Sarvazyan et al., 2024) achieved the highest performance in this subtask by extracting token-level probabilistic fea-

tures (log probability and entropy) using four LLaMA-2 models: LLaMA-2-7B, LLaMA-2-7B-chat, LLaMA-2-13B, and LLaMA-2-13B-chat. These features were then fed into a Transformer Encoder trained in a supervised manner.

Team USTC-BUPT_{STA_mono:2} (Guo et al., 2024) secured the second position. Their model is built upon RoBERTa, with the addition of two classification heads: one for binary classification (human or machine) using MLP layers, and another for domain classification (e.g., news, essays, etc.). The latter is equipped with an MLP layer and a gradient reversal layer to enhance transferability between the training and test sets. A sum-up loss is applied, resulting in approximately 8% improvement compared to the RoBERTa baseline.

Team PetKaz_{STA_mono:12} (Petukhova et al., 2024) utilized a fine-tuned RoBERTa augmented with diverse linguistic features.

In addition to the top three teams: **Team HU_{STA_mono:17}** (Roy Dipta and Shahriar, 2024) employed a contrastive learning-based approach, fine-tuning MPNet on an augmented dataset. **Team TrustAI_{STA_mono:20}** ensembles several classical ML classifiers, Naive Bayes, LightGBM and SGD. **Team L3i++_{STA_mono:24}** (Tran et al., 2024) investigated various approaches including likelihood, fine-tuning small PLMs, and LLMs, with the latter, fine-tuned LLaMA-2-7B, proving to be the most effective. **Team art-nat-HHU_{STA_mono:25}** (Ciccarelli et al., 2024) utilized a RoBERTa-base model combined with syntactic, lexical, probabilistic, and stylistic features. **Team Unibuc - NLP_{STA_mono:28}** (Marchitan et al., 2024) jointly trained Subtasks A and B based on RoBERTa. Most other teams fine-tuned either RoBERTa or XLM-RoBERTa for MGT detection, enhancing the models through various techniques, ranging from a mixture of experts by **Team Werkzeug_{STA_mono:45}** (Wu et al., 2024) to low-rank adaptation by **Team NCL-UoR_{STA_mono:50}** (Xiong et al., 2024), while **Team Sharif-MGTD_{STA_mono:51}** (Ebrahimi et al., 2024) preferred careful fine-tuning of PLMs alone.

5.2 Multilingual Human vs Machine

Table 6 provides an overview of the methods employed by the top-performing systems for Subtask A Multilingual. Various techniques are utilized, including zero-shot learning based on LLMs, PLM-based classifiers, and ensemble models.

Team USTC-BUPT_{STA_Multi:1} (Guo et al.,

Team Name	Ranking	small PLM	LLM	GPT	Fine-tuning	Zero-shot	Data augmentation	External Data
USTC-BUPT	1		✓		✓			
FI Group	2	✓			✓			
KInIT	3		✓		✓	✓		
L3i++	5		✓		✓			
QUST	6		✓		✓		✓	
AIpom	9		✓		✓			
SINAI	21		✓		✓			
Unibuc-NLP	22	✓			✓			
Werkzeug	30	✓			✓			
RUG-5	32	✓	✓		✓			
DUTh	33		✓		✓			
RUG-D	39		✓		✓			✓
MasonTigers	49	✓	✓		✓	✓		
TrustAI	55	✓			✓			

Table 6: Subtask A multilingual participants methods.

2024) secured the top position. They initially detect the language of the input text. For English text, they average embeddings from LLaMA-2-70B, followed by classification through a two-stage CNN. For texts in other languages, the classification problem is transformed into fine-tuning a next-token prediction task using the mT5 model, incorporating special tokens for classification. Their approach integrates both monolingual and multilingual strategies, exploiting large language models for direct embedding extraction and model fine-tuning. This enables the system to adeptly handle text classification across a diverse range of languages, especially those with fewer resources.

Team FI Group_{STA_Multi:2} (Ben-Fares et al., 2024) implemented a hierarchical fusion strategy that adaptively combines representations from different layers of XLM-RoBERTa-large, moving beyond the conventional "[CLS]" token classification to sequence labeling for enhanced detection of stylistic nuances.

Team KInIT_{STA_Multi:3} (Spiegel and Macko, 2024) combined fine-tuned LLMs with zero-shot statistical methods, employing a two-step majority voting system for predictions. Their method emphasizes language identification, per-language threshold calibration, and the integration of both fine-tuned and statistical detection methods, demonstrating the power of ensemble strategies. For the LLMs, they utilized QLoRA PEFT to fine-tune

Team Name	Ranking	small PLM	LLM	GPT	fine-tuning	zero-shot	Data augmentation
AISPACE	1		✓		✓		✓
Unibuc - NLP	2	✓			✓		✓
USTC-BUPT	3		✓				
L3i++	6		✓		✓		
MLab	7				✓		
Werkzeug	8	✓			✓		
TrustAI	14	✓			✓		
MGTD4ADL	17				✓		✓
scalar	18	✓					✓
UMUT	23	✓			✓		
QUST	36		✓		✓		✓
MasonTigers	38	✓	✓		✓	✓	
RUG-5	41	✓	✓		✓		
RUG-D	44		✓		✓		
DUTh	49		✓		✓		
clulab-UofA	62		✓	✓	✓		✓

Table 7: **Subtask B** Participants method overview.

Falcon-7B and Mistral-7B.

Other teams explored various approaches, like using LoRA-finetuned LLMs as classifiers (**Team AIpom_{STA_Multi:9}**) (Shirnin et al., 2024), using semantic and syntactic aspects of the texts (**RFBS_{STA_Multi:10}**) (Heydari Rad et al., 2024) or fusing perplexity with text and adding a classification head (**Team SINAI_{STA_Multi:21}**) (Gutiérrez Megías et al., 2024). Each team’s method provides insights into the complexities of multilingual text detection, ranging from the use of specific LLMs and PLMs to the use of linguistic and probabilistic metrics and ensemble techniques (Wu et al., 2024; Brekhof et al., 2024; Kyriakou et al., 2024; Puspo et al., 2024; Urlana et al., 2024).

5.3 Multi-way Detection

Table 7 provides an overview of the approaches employed by the top-ranking systems for Subtask B. Similar to Subtask A, most solutions do not use GPT and zero-shot approaches. The best-performing solutions primarily exploit LLMs and data augmentation.

Team AISPACE_{STB:1} (Gu and Meng, 2024) achieved the highest performance in this subtask by fine-tuning various encoder and encoder-decoder

models, including RoBERTa, DeBERTa, XLNet, Longformer, and T5. They augmented the data with instances from Subtask A and explored the effects of different loss functions and learning rate values. Their method leverages a weighted Cross-Entropy loss to balance samples in different classes and uses an ensemble of fine-tuned models to improve robustness.

Team Unibuc - NLP_{STB:2} (Marchitan et al., 2024) employed a Transformer-based model with a unique two-layer feed-forward network as a classification head. They also augmented the data with instances from the Subtask A monolingual dataset.

Team USTC-BUPT_{STB:3} (Guo et al., 2024) leveraged LLaMA-2-70B to obtain token embeddings and applied a three-stage classification. They first distinguished human-generated from machine-generated text using LLaMA-2-70B, then categorized ChatGPT and Cohere as one class and distinguished them from davinci-003, BLOOMz, and Dolly-v2. Finally, they performed binary classification between ChatGPT and Cohere.

Team L3i++_{STB:6} (Tran et al., 2024) conducted a comparative study among three groups of methods: metric-based models, fine-tuned classification language models (RoBERTa, XLM-R), and a fine-tuned LLM, LLaMA-2-7B, finding LLaMA-2 to outperform other methods. They analyzed errors and various factors in their paper.

Team MLab_{STB:7} (Li et al., 2024) fine-tuned DeBERTa and analyzed the embeddings from the last layer to provide insights into the embedding space of the model.

Team Werkzeug_{STB:8} (Wu et al., 2024) utilized RoBERTa-large and XLM-RoBERTa-large to encode text, addressing the problem of anisotropy in text embeddings produced by pre-trained language models (PLMs) by introducing a learnable parametric whitening (PW) transformation. They also used multiple PW transformation layers as experts under the mixture-of-experts (MoE) architecture to capture features of LLM-generated text from different perspectives.

Other teams explored various approaches, including different loss functions and sentence transformers (**Team MGTD4ADL_{STB:17}**) (Chen et al., 2024), RoBERTa fine-tuning (**Team UMUTeam_{STB:23}**) (pan et al., 2024), stacking ensemble techniques (**Team MasonTigers_{STB:38}**) (Puspo et al., 2024), and basic ML models with linguistic-stylistic features (**Team RUG-5_{STB:41}**)

Team Name	Ranking	small PLM	LLM	LSTM (+) CNN	fine-tuning	Data augmentation	CRF layer
TM-TREK _{STC:1}	1	✓			✓		✓
AIpom _{STC:2}	2		✓		✓		
USTC-BUPT _{STC:3}	3	✓	✓		✓	✓	
RKadiyala _{STC:6}	6	✓			✓		✓
DeepPavlov _{STC:7}	7	✓			✓	✓	
RUG-5 _{STC:17}	17	✓	✓		✓		
TueCICL _{STC:22}	22			✓			
jelarson _{STC:25}	25						
MasonTigers _{STC:27}	27	✓					
Unibuc-NLP _{STC:28}	28			✓	✓		

Table 8: **Subtask C** Participants method overview.

(Darwinkel et al., 2024).

5.4 Boundary Identification

Table 8 presents an overview of the methods used by the top-ranking systems for Subtask C. The best performing solutions are mainly based on ensemble strategies, with some employing data augmentation.

Team TM-TREK_{STC:1} (Qu and Meng, 2024) achieved the highest performance in Subtask C. They utilized an ensemble of XLNet models, each trained with a distinct seed, and used a straightforward voting mechanism on the output logits. They also explored the integration of LSTM and CRF layers on top of various PLMs, along with continued pretraining and fine-tuning of PLMs, and dice loss functions to enhance model performance.

Team AIpom_{STC:2} (Shirnin et al., 2024) introduced a novel two-stage pipeline merging outputs from an instruction-tuned, decoder-only (Mistral-7B-OpenOrca) model and two encoder-only sequence taggers.

Team USTC-BUPT_{STC:3} (Guo et al., 2024) fine-tuned a DeBERTa model with data augmentation and framed the task as a token classification problem.

Team RKadiyala_{STC:6} (Kadiyala, 2024) fine-tuned various encoder-based models with a Conditional Random Field (CRF) layer and found DeBERTa-V3 to perform the best on the development set.

Team DeepPavlov_{STC:7} (Voznyuk and Konovalov, 2024) fine-tuned the DeBERTa-v3 model using the provided dataset and developed a data prepro-

cessing pipeline for data augmentation.

Other teams explored diverse CNN, LSTM (Team TueCICL_{STC:22}) (Stuhlinger and Winkler, 2024), (Team Unibuc - NLP_{STC:28}) (Marchitan et al., 2024), and regression-based (Team jelarson_{STC:25}) (Larson and Tyers, 2024) techniques to address this challenge, although many did not surpass the baselines due to issues related to model overfitting or inadequate word embeddings.

6 Results and Discussion

6.1 Subtask A

There were three submissions for subtask A, which were submitted in time, but had the wrong file name, which prevented us from scoring them automatically. We eventually manually fixed the names and scored them, and we also added them to the ranking but marked them with a *. They should be considered as unofficial submissions.

Monolingual Table 9 presents the performance of systems in the monolingual track of Subtask A. Out of 125 participating teams, 15 surpassed the baseline, with the top-performing team (Genaïos) achieving an accuracy of 96.88. Notably, several teams demonstrated high precision and recall scores, indicating robust performance in distinguishing between human-generated and machine-generated text in a binary classification context.

Multilingual Table 10 presents the performance of systems in the multilingual track of Subtask A, where Team USTC-BUPT emerges as the top performer among 62 participating teams, achieving an accuracy of 95.99, remarkably close to the English-only result. Their methodology entails a blend of language detection and fine-tuning tasks using LLaMA-2-70B for English and the mT5 model for others, showcasing their adaptability across diverse languages.

Similarly, among the 22 teams surpassing the baseline, the majority leverage advanced LLMs such as LLaMA, Mistral, etc., while also emphasizing syntax and writing style differences between human and AI-generated texts. For example, Team FI Group implements a hierarchical fusion strategy to adaptively fuse representations from different BERT layers, prioritizing syntax over semantics for improved classification accuracy. Likewise, Team KInIT employs an ensemble approach, combining fine-tuned LLMs with zero-shot statistical methods,

Rank	Team	Prec	Recall	F1-score	Acc	Rank	Team	Prec	Recall	F1-score	Acc
*	dianchi	96.21	99.19	97.68	97.53	62	Collectivized Semantics	68.21	99.39	80.90	75.35
1	Genaios	96.11	98.03	97.06	96.88	63	IUCL	68.13	98.33	80.49	74.96
2	USTC-BUPT	95.75	96.86	96.30	96.10	64	annedadaa	68.01	97.69	80.19	74.66
3	mail6djj	94.87	97.18	96.02	95.76	65	cmly99	67.92	97.96	80.22	74.62
4	howudoin	93.48	98.12	95.74	95.42	66	xiaoll	67.92	97.96	80.22	74.62
5	idontknow	94.57	95.42	94.99	94.72	67	SINAI	67.31	99.88	80.42	74.46
6	seven	90.12	98.31	94.04	93.46	68	yuwert777	68.78	92.96	79.06	74.14
7	zongxiong	93.54	93.82	93.68	93.35	69	yaoxy	68.78	92.96	79.06	74.14
8	mahsaamani	90.59	96.23	93.32	92.77	70	moniszc	67.25	97.66	79.65	73.79
9	bennben	91.49	95.05	93.24	92.76	71	MasonTigers	67.59	95.72	79.23	73.64
10	infinity2357	91.92	90.96	91.43	91.05	72	AT	67.30	96.59	79.33	73.56
11	AISPACE	84.76	99.92	91.72	90.52	73	DUT	66.27	99.92	79.69	73.24
12	petkaz	85.54	98.59	91.61	90.51	74	surbhi	69.38	87.40	77.35	73.12
13	moniszc1	86.96	95.68	91.11	90.20	75	thanet	69.47	86.69	77.13	73.00
14	moniszc3	86.96	95.68	91.11	90.20	76	Kathlalu	74.47	73.89	74.18	72.98
15	flash	82.39	99.77	90.25	88.68	77	KInIT	66.14	98.44	79.12	72.71
*	baseline	93.36	84.02	88.44	88.47	78	iimasNLP	67.81	87.08	76.25	71.50
16	ericmxf	81.71	99.98	89.93	88.24	79	wwzzhh	64.38	99.49	78.18	70.82
17	HU	82.63	97.24	89.34	87.81	80	bharathsk	64.48	98.69	78.00	70.76
18	jrutkowski2	84.58	93.44	88.79	87.61	81	apillay2	64.48	98.69	78.00	70.76
19	lihaoran	89.26	85.86	87.53	87.15	82	ashinee20	71.91	71.57	71.74	70.39
20	TrustAI	89.21	85.50	87.31	86.95	83	longfarmer	63.81	99.03	77.61	69.99
21	TM-TREK	79.47	99.99	88.56	86.43	84	mlnick	63.61	100	77.76	69.96
22	jojoc	86.30	87.76	87.02	86.25	85	vasko	63.47	99.93	77.63	69.75
23	RFBS	91.58	80.64	85.76	85.95	86	Groningen F	72.74	65.62	68.99	69.02
24	L3i++	81.41	94.66	87.53	85.84	87	hhy123	62.76	99.94	77.11	68.83
25	art-nat-HHU	86.29	86.04	86.17	85.49	88	1024m	62.54	99.98	76.95	68.53
26	FI Group	79.52	96.99	87.39	85.30	89	lhy123	62.54	99.96	76.94	68.53
27	phuhoang	87.72	83.65	85.64	85.26	90	thang	62.25	99.98	76.73	68.14
28	Unibuc-NLP	78.01	99.86	87.59	85.14	91	nikich28	61.90	99.26	76.25	67.52
29	sushvin	82.65	89.76	86.06	84.73	92	niceone	61.70	98.39	75.84	67.08
30	NewbieML	79.32	95.06	86.48	84.39	93	pmalea	60.73	97.87	74.95	65.64
31	QUST	76.88	99.91	86.89	84.17	94	mahaalblooki	60.85	96.23	74.56	65.51
32	MLab	83.17	85.83	84.48	83.44	95	bertsquad	60.32	98.94	74.95	65.27
33	ziweizheng	82.31	85.01	83.63	82.53	96	jjonczyk	60.25	99.31	75.00	65.23
34	Alpom	74.34	99.97	85.27	81.86	97	dkoterwa	60.12	99.98	75.09	65.16
35	lyaleo	79.38	87.82	83.39	81.62	98	lystsoval	92.30	35.31	51.07	64.48
36	yunhfang	75.26	95.31	84.10	81.08	99	sunilgundapu	59.22	99.31	74.20	63.72
37	sankalpabhad	78.22	88.08	82.86	80.86	100	RUG-D	59.19	99.35	74.18	63.68
38	aktsvigun	78.22	88.08	82.86	80.86	101	RUG-5	60.79	84.13	70.58	63.17
39	NootNoot	78.22	88.08	82.86	80.86	102	harshul24	54.55	57.14	55.81	62.00
40	Mast Kalandar	74.65	96.16	84.05	80.83	103	basavraj10	54.55	57.14	55.81	62.00
41	I2C-Huelva	73.92	98.01	84.28	80.79	104	samnlptaskab	58.01	99.93	73.41	61.97
42	priority497	73.31	99.69	84.49	80.78	105	partnlu	57.87	99.96	73.31	61.76
43	wjm123	73.31	99.69	84.49	80.78	106	teams2024	57.78	99.97	73.23	61.61
44	scalar	73.10	99.97	84.45	80.67	107	Rkadiyala	57.30	99.98	72.85	60.86
45	werkzeug	75.28	93.88	83.56	80.59	108	rtuora	57.18	99.89	72.73	60.66
46	blain	72.51	99.96	84.05	80.07	109	teamlanlp2	56.93	99.71	72.48	60.24
47	xxm981215	72.32	99.79	83.86	79.83	110	jakubbebac	56.89	99.88	72.49	60.18
48	moyanxinxu	72.32	99.79	83.86	79.83	111	dandread	55.19	99.87	71.92	59.04
49	jrutkowskikag1	73.02	97.54	83.52	79.78	112	pask1	55.97	99.80	71.72	58.67
50	NCL-UoR	75.10	90.84	82.22	79.37	113	skillissue	55.14	100	71.09	57.27
51	Sharif-MGTD	73.41	93.75	82.35	78.89	114	RUG-3	54.92	99.73	70.83	56.87
52	wgm123	71.16	99.94	83.13	78.69	115	Mashee	57.11	59.58	58.32	55.27
53	logiczmaksimka	70.79	99.29	82.65	78.11	116	TueCICL	55.37	69.61	61.68	54.57
54	somerandomjj	70.43	98.55	82.15	77.51	117	RUG-1	52.52	100	68.87	52.52
55	totylkokuba	70.43	98.55	82.15	77.51	118	novice8	52.24	68.68	59.34	50.57
56	lly123	69.25	99.95	81.81	76.66	119	ronghaopan	52.49	38.47	44.40	48.70
57	mimkag2	69.69	97.99	81.45	76.56	120	kamer	52.89	29.56	37.93	48.48
58	priyansk	69.28	99.36	81.64	76.53	121	helenpy	75.29	1.79	3.51	48.11
59	nampfiey1995	77.92	76.93	77.43	76.44	122	ascisel	7.14	0.01	0.01	47.44
60	xiangrunli	68.23	99.97	81.11	75.54	123	laida	40.18	19.28	26.06	42.53
61	roywang	68.23	99.97	81.11	75.54	124	nz28555	40.31	33.18	36.40	39.10
*	badrock	71.13	89.50	79.27	75.41						

Table 9: **Subtask A monolingual** Prec (precision), Recall, and F1-scores(%) with respect to **MGT**.

Rank	Team	Prec	Recall	F1-score	Acc
1	USTC-BUPT	94.93	97.53	96.21	95.99
2	FI Group	94.28	98.00	96.10	95.85
3	KInIT	92.95	97.86	95.34	95.00
4	priyansk	90.70	98.14	94.28	93.77
5	L3i++	92.47	94.00	93.23	92.87
6	QUST	90.45	90.98	90.71	90.27
7	xxm981215	90.45	90.98	90.71	90.27
8	NCL-UoR	81.42	95.41	87.86	86.23
9	AIpom	80.72	95.80	87.61	85.85
10	RFBES	85.43	85.27	85.35	84.71
11	blain	76.12	98.67	85.94	83.14
12	xiangrunli	75.20	99.67	85.73	82.66
13	wgm123	75.20	99.67	85.73	82.66
14	roywang	75.08	99.75	85.68	82.58
15	logiczmaaksimka	74.34	99.33	85.04	81.74
16	zaratiana	74.75	96.68	84.31	81.21
17	thanet	76.18	92.56	83.58	81.00
*	baseline	73.45	99.30	84.44	80.89
18	cmy99	73.29	99.61	84.45	80.83
19	lly123	73.09	99.67	84.33	80.65
20	moyanxinxu	73.09	99.67	84.33	80.65
21	SINAI	72.51	99.91	84.04	80.17
22	Unibuc-NLP	71.82	99.79	83.52	79.43
23	annedadaa	72.16	98.57	83.32	79.39
24	1024m	71.03	99.91	83.03	78.66
25	sunilgundapu	71.04	98.86	82.67	78.35
26	hirak	70.79	99.66	82.78	78.34
27	bertsquad	70.45	99.12	82.36	77.82
28	Rkadiyala	69.99	99.94	82.33	77.59
*	dianchi	69.88	99.91	82.24	77.46
29	lyaleo	69.50	99.83	81.95	77.03
30	werkzeug	69.33	99.81	81.82	76.83
31	mlnick	69.25	99.81	81.77	76.75
32	RUG-5	69.90	96.78	81.17	76.55
33	DUTh	68.95	99.93	81.60	76.45
34	dandread	68.31	99.75	81.09	75.69
35	Genaios	68.30	99.73	81.07	75.67
36	vasko	67.99	98.68	80.50	75.03
37	thang	67.16	99.78	80.29	74.40
38	mahsaamani	68.53	93.21	78.98	74.09
39	RUG-D	65.03	99.55	78.67	71.79
40	omarnasr	64.80	99.21	78.40	71.43
41	lhy123	64.62	99.85	78.46	71.36
42	priority497	64.62	99.85	78.46	71.36
43	hhy123	64.47	99.84	78.35	71.17
44	wjm123	64.47	99.84	78.35	71.17
45	aktsvigun	62.83	99.36	76.98	68.96
46	sankalpbad	62.83	99.36	76.98	68.96
47	NootNoot	62.83	99.36	76.98	68.96
48	nampfiev1995	61.37	77.15	68.36	62.70
49	MasonTigers	56.77	100	72.42	60.21
50	RUG-1	51.33	97.39	67.23	51.22
51	novice8	51.95	84.56	64.36	51.08
52	scalar	52.04	80.17	63.11	51.04
53	mahaalblooki	48.96	51.24	50.08	50.55
54	Sharif-MGTD	51.42	67.13	58.23	50.53
55	TrustAI	51.15	62.04	56.07	50.06
56	sky2024just	51.96	26.37	34.99	48.79
57	laida	48.88	20.56	28.94	47.27

Table 10: **Subtask A multilingual** Prec (precision), Recall, and F1-scores(%) with respect to MGT.

Rank	Team	Prec	Recall	F1-score	Acc
1	AISPACE	91.81	90.85	90.84	90.85
2	Unibuc - NLP	88.69	86.96	87.03	86.96
3	USTC-BUPT	89.54	84.33	82.72	84.33
4	dianchi	86.45	83.48	83.62	83.48
5	NootNoot	86.68	83.12	83.15	83.12
6	L3i++	86.01	83.12	83.08	83.12
7	MLab	85.00	82.67	82.76	82.67
8	werkzeug	86.30	82.23	81.63	82.23
9	flash	88.29	82.23	79.46	82.23
10	juse7198	86.83	82.03	80.72	82.03
11	idontknow	88.44	80.94	77.47	80.94
12	TM-TREK	86.42	79.84	79.46	79.84
13	howudoin	80.02	79.68	79.79	79.68
14	TrustAI	83.80	79.19	79.07	79.19
15	I2C-Huelva	84.45	78.90	78.82	78.90
16	ericmxf	85.52	78.74	76.88	78.74
17	MGTD4ADL	83.78	76.96	74.46	76.96
18	scalar	81.90	76.26	76.00	76.26
19	ronghaopan	81.11	75.19	71.38	75.35
20	sunilgundapu	81.06	75.06	73.81	75.06
*	baseline	81.14	74.61	72.59	74.61
21	Collectivized Semantics	82.35	73.87	70.26	73.87
22	priyansk	78.06	73.36	67.05	73.36
23	logiczmaaksimka	67.73	69.13	64.36	69.13
24	annedadaa	79.55	68.98	64.55	68.98
25	hhy123	65.94	67.77	63.01	67.77
26	xiangrunli	65.94	67.77	63.01	67.77
27	wjm123	65.94	67.77	63.01	67.77
28	lhy123	65.94	67.77	63.01	67.77
29	lly123	65.94	67.77	63.01	67.77
30	wgm123	65.94	67.77	63.01	67.77
31	moyanxinxu	65.94	67.77	63.01	67.77
32	priority497	65.94	67.77	63.01	67.77
33	thang	66.36	67.68	63.79	67.68
34	blain	63.15	67.23	62.35	67.23
35	xxm981215	65.77	67.21	62.41	67.21
36	QUST	65.77	67.21	62.41	67.21
37	mahaalblooki	63.72	66.27	61.82	66.27
38	MasonTigers	73.62	65.04	64.47	65.04
39	Rkadiyala	65.81	64.91	59.98	64.91
40	1024m	66.10	64.38	59.82	64.38
41	RUG-5	62.21	64.21	59.04	64.21
42	thanet	63.42	61.88	55.58	61.88
43	mlnick	66.84	61.79	57.53	61.79
44	RUG-D	66.39	61.54	53.82	61.54
45	Groningen F	60.10	60.84	57.90	60.84
46	NCL-UoR	69.03	60.15	58.05	60.15
47	mahsaamani	60.41	59.42	52.89	59.42
48	dandread	71.95	58.35	52.28	58.35
49	DUTh	63.71	56.68	51.25	56.68
50	bertsquad	57.27	55.97	51.49	55.97
51	RUG-3	61.51	54.23	49.26	54.23
52	cmy99	58.04	53.35	50.58	53.35
53	skysky12	60.86	53.31	50.14	53.31
54	vasko	59.98	52.82	50.38	52.82
55	phuhoang	61.54	50.79	50.21	50.79
56	rtuora	54.21	50.32	44.15	50.32
57	AT	53.13	48.59	43.91	48.59
58	teams2024	45.50	47.01	41.05	47.01
59	windwind22	39.87	39.31	32.79	39.31
60	helenpy	39.88	38.27	32.20	38.27
61	iimasNLP	39.88	38.27	32.20	38.27
62	clulab-UofA	37.53	29.29	24.58	29.29
63	samnlptaskab	25.78	27.81	21.07	27.81
64	mhr2004	17.06	17.06	17.06	17.06
65	xiaoll	5.73	17.15	8.47	17.02
66	surbhi	17.24	16.77	15.10	16.77
67	roywang	2.78	16.67	4.76	16.67
68	RUG-1	2.78	16.67	4.76	16.67
69	novice8	16.39	16.55	13.93	16.55
70	NewbieML	15.99	15.58	14.13	15.30

Table 11: **Subtask B: Multi-Way Generator Detection** Prec (precision), Recall, and F1-scores(%) macro average.

resulting in a unique combination of techniques that effectively enhances classification accuracy.

Overall, these successful methodologies under-

Rank	Team	MAE	Rank	Team	MAE
1	TM-TREK	15.68	16	mahaalblooki	25.95
2	Alpom	15.94	17	RUG-5	26.07
3	USTC-BUPT	17.70	18	mahsaamani	26.27
4	ywnh111	18.08	19	aktsvigun	26.40
5	ywnh222	18.51	20	skillissue	27.99
6	Rkadiyala	18.54	21	NootNoot	28.01
7	DeepPavlov	19.25	22	TueCICL	34.88
8	knk42	19.42	23	dandread	35.17
9	vasko	19.93	24	novice8	44.82
10	logiczmaksimka	19.93	25	jelaron	48.14
11	AISPACE	21.19	26	TueSents	58.95
*	baseline	21.54	27	MasonTigers	60.78
12	ericmxf	21.55	28	Unibuc - NLP	74.28
13	blain	21.80	29	lanilequiu	78.18
14	1024m	22.36	30	scalar	87.72
15	cmy99	24.68			

Table 12: **Subtask C: Boundary Identification.**

score the importance of leveraging advanced LLMs, ensemble techniques, and comprehensive analysis to achieve superior performance in detecting machine-generated text across multilingual contexts.

6.2 Subtask B

For Subtask B (Multi-Way detection), 70 teams participated, with 20 surpassing the baseline of 74.61 accuracy. Table 11 displays the full results. In summary, the subtask results underline the effectiveness of diverse and innovative approaches, including fine-tuning advanced models (e.g., RoBERTa, DeBERTa, XLNet, Longformer, T5), data augmentation (e.g., using Subtask A instances), ensemble strategies, and the exploration of novel loss functions and learning techniques. The leading entries showcased a range of methodologies, from leveraging the power of large language models and addressing embedding anisotropy to integrating traditional and neural methods, underscoring the dynamic and evolving nature of NLP research. For instance, Team AISPACE utilized a weighted Cross-Entropy loss and an ensemble approach based on model performance per class, which led to the highest accuracy of 90.85.

6.3 Subtask C

Of the 30 systems that were submitted for Subtask C, 11 outperformed the baseline MAE of 21.54. The top system, TM-TREK, achieved the best submitted MAE of 15.68. A significant majority of the top-performing teams relied on ensembles of PLMs, indicating a consensus that combining the strengths of multiple models can lead to more robust and accurate predictions. This approach lever-

ages the diverse representations and strengths of different models to mitigate weaknesses inherent in individual systems.

Data augmentation emerged as a critical strategy among leading teams, suggesting its effectiveness in enhancing model performance by providing a richer, more varied training dataset. This includes both the generation of new training examples and the manipulation of existing data to better capture the complexity and variability of natural language.

Despite the advanced methodologies deployed, some teams struggled with issues related to overfitting and the adequacy of word embeddings. This underscores the ongoing challenges in developing models that generalize well to unseen data and the critical role of embeddings in capturing semantic and syntactic nuances of language.

7 Conclusion and Future Work

We have described SemEval-2024 Task 8 on Multi-generator, Multidomain, and Multilingual Machine-Generated Text Detection. The task garnered significant interest from researchers, with 126, 59, 70, and 30 teams submitting entries for Subtask A Monolingual, Subtask A Multilingual, Subtask B, and Subtask C, respectively. Additionally, we received 54 system description papers before finalizing this submission.

Overall, Subtasks A and B were relatively easier, with all systems showing improvements over the baseline. However, Subtask C proved to be significantly more challenging. Fewer teams participated, and many struggled to surpass our baseline results set in (Wang et al., 2024a).

In future work, we plan to extend our focus beyond machine-generated text detection to other modalities such as image, speech, and video detection. Additionally, we intend to develop an open-source demonstration system capable of distinguishing between AI-generated content and human-produced content.

Limitations

Despite providing a comprehensive dataset that spans multiple languages, generators, and domains across three distinct tasks in machine-generated text detection, our study encounters several limitations that pave the way for future research.

Firstly, the reliance on textual data without access to white-box information, such as token-level probabilities, confines our detection methods to

black-box approaches across all tasks. These methods might exhibit reduced effectiveness and struggle to generalize across new domains, generators, and languages. Additionally, they are susceptible to language-style attacks, including paraphrasing in different tones, back-translation, and other forms of textual adversarial tactics. In contrast, methods that leverage watermarking and white-box patterns show greater promise for robust MGT detection.

Secondly, our approach to boundary identification presupposes that each text comprises an initial segment written by humans followed by machine-generated content, with only one transition point. However, real-world scenarios often present more complex challenges. It is crucial not only to ascertain the presence of mixed text but also to identify all transition points. Texts may originate from human authors and undergo refinement via machine assistance, or vice versa, encompassing machine generation followed by human revision. Addressing these nuanced scenarios will be a focus of our future research efforts.

Ethics and Broader Impact

This section outlines potential ethical considerations related to our work.

Data Collection and Licenses Our study utilizes pre-existing corpora, specifically the M4 and OUTFOX datasets, which have been publicly released for research purposes under clear licensing agreements.

Security Implications The dataset underpinning our shared task aims to foster the development of robust MGT detection systems. These systems are crucial for identifying and mitigating misuse scenarios, such as curbing the proliferation of automated misinformation campaigns and protecting individuals and institutions from potential financial losses. In fields such as journalism, academia, and legal proceedings, where the authenticity of information is of utmost importance, MGT detection plays a vital role in maintaining content integrity and trust. Furthermore, by enhancing public awareness of the capabilities and limitations of LLMs, we can cultivate a healthy skepticism towards digital content. Effective MGT detection mechanisms are essential for ensuring that users can place their trust in content generated by LLMs.

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References

- Pranjal Aggarwal and Deepanshu Sachdeva. 2024. [Cunlp at semeval-2024 task 8: Classify human and ai generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1–6, Mexico City, Mexico. Association for Computational Linguistics.
- Huseyin Alecakir, Puja Chakraborty, Pontus Henningsson, Matthijs van Hofslot, and Alon Scheuer. 2024. [Groningen team a at semeval-2024 task 8: Human/machine authorship attribution using a combination of probabilistic and linguistic features](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1931–1937, Mexico City, Mexico. Association for Computational Linguistics.
- Jainit Bafna, Hardik Mittal, Suyash Sethia, Manish Shrivastava, and Radhika Mamidi. 2024. [Mast kalandar at semeval-2024 task 8: On the trail of textual origins: Roberta-bilstm approach to detect ai-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1638–1644, Mexico City, Mexico. Association for Computational Linguistics.
- Sankalp Bahad, Yash Bhaskar, and Parameswari Krishnamurthy. 2024. [Fine-tuning language models for ai vs human generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 905–908, Mexico City, Mexico. Association for Computational Linguistics.
- Maha Ben-Fares, Urchade Zaratiana, Simon Hernandez, and Pierre Holat. 2024. [Fi group at semeval-2024 task 8: A syntactically motivated architecture for multilingual machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1155–1160, Mexico City, Mexico. Association for Computational Linguistics.
- Thijs Brekhof, Xuanyi Liu, Yuwen Zhou, and Joris Ruitenbeek. 2024. [Groningen team d at semeval-2024 task 8: Exploring data generation and a combined model for fine-tuning llms for multidomain machine-generated text detection](#). In *Proceedings of*

- the 18th International Workshop on Semantic Evaluation (SemEval-2024), pages 378–385, Mexico City, Mexico. Association for Computational Linguistics.
- Lujia Cao, Ece Lara Kilic, and Katharina Will. 2024. [Kathlalu at semeval-2024 task 8: A comparative analysis of binary classification methods for distinguishing between human and machine-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 386–389, Mexico City, Mexico. Association for Computational Linguistics.
- Huixin Chen, Jan Büssing, David Rügamer, and Ercong Nie. 2024. [Team mgtd4adl at semeval-2024 task 8: Leveraging \(sentence\) transformer models with contrastive learning for identifying machine-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1722–1729, Mexico City, Mexico. Association for Computational Linguistics.
- Vittorio Ciccarelli, Cornelia Genz, Nele Mastracchio, Wiebke Petersen, Anna Stein, and Hanxin Xia. 2024. [Team art-nat-hhu at semeval-2024 task 8: Stylistically informed fusion model for mgt-detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1701–1708, Mexico City, Mexico. Association for Computational Linguistics.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Unsupervised cross-lingual representation learning at scale. *arXiv preprint arXiv:1911.02116*.
- Evan Crothers, Nathalie Japkowicz, Herna Viktor, and Paula Branco. 2022. Adversarial robustness of neural-statistical features in detection of generative transformers. In *2022 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE.
- Evan Crothers, Nathalie Japkowicz, and Herna L Viktor. 2023. Machine-generated text: A comprehensive survey of threat models and detection methods. *IEEE Access*.
- Patrick Darwinkel, Sijbren van Vaals, Marieke van der Holt, and Jarno van Houten. 2024. [Groningen group e at semeval-2024 task 8: Detecting machine-generated texts through pre-trained language models augmented with explicit linguistic-stylistic features](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 995–1003, Mexico City, Mexico. Association for Computational Linguistics.
- Ayan Datta, Aryan Chandramania, and Radhika Mamidi. 2024. [Weighted layer averaging roberta for black-box machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1634–1637, Mexico City, Mexico. Association for Computational Linguistics.
- Rina Donker, Björn Overbeek, Dennis van Thulden, and Oscar Zwagers. 2024. [Groningen team f at semeval-2024 task 8: Detecting machine-generated text using feature-based machine learning models](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1924–1930, Mexico City, Mexico. Association for Computational Linguistics.
- Liam Dugan, Daphne Ippolito, Arun Kirubakaran, Sherry Shi, and Chris Callison-Burch. 2023. Real or fake text?: Investigating human ability to detect boundaries between human-written and machine-generated text. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 12763–12771.
- Seyedeh Fatemeh Ebrahimi, Karim Akhavan Azari, Amirmasoud Irvani, Arian Qazvini, Pouya Sadeghi, Zeinab Taghavi, and Hossein Sameti. 2024. [Sharif-mgtd at semeval-2024 task 8: A transformer-based approach to detect machine generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 552–559, Mexico City, Mexico. Association for Computational Linguistics.
- Chujie Gao, Dongping Chen, Qihui Zhang, Yue Huang, Yao Wan, and Lichao Sun. 2024. Llm-as-a-coauthor: The challenges of detecting llm-human mixcase. *arXiv preprint arXiv:2401.05952*.
- Sebastian Gehrmann, Hendrik Strobelt, and Alexander Rush. 2019a. [GLTR: Statistical detection and visualization of generated text](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 111–116, Florence, Italy. Association for Computational Linguistics.
- Sebastian Gehrmann, Hendrik Strobelt, and Alexander M Rush. 2019b. Gltr: Statistical detection and visualization of generated text. *arXiv preprint arXiv:1906.04043*.
- Katy Ilonka Gero, Vivian Liu, and Lydia Chilton. 2022. Sparks: Inspiration for science writing using language models. In *Designing interactive systems conference*, pages 1002–1019.
- Renhua Gu and Xiangfeng Meng. 2024. [Aispace at semeval-2024 task 8: A class-balanced soft-voting system for detecting multi-generator machine-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1487–1492, Mexico City, Mexico. Association for Computational Linguistics.
- Biyang Guo, Xin Zhang, Ziyuan Wang, Minqi Jiang, Jinran Nie, Yuxuan Ding, Jianwei Yue, and Yupeng Wu. 2023. [How close is chatgpt to human experts? comparison corpus, evaluation, and detection](#). *CoRR*, abs/2301.07597.

- Zikang Guo, Kaijie Jiao, Xingyu Yao, Yuning Wan, Haoran Li, Benfeng Xu, Licheng Zhang, Quan Wang, Yongdong Zhang, and Zhendong Mao. 2024. [Ustc-bupt at semeval-2024 task 8: Enhancing machine-generated text detection via domain adversarial neural networks and llm embeddings](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1522–1533, Mexico City, Mexico. Association for Computational Linguistics.
- Alberto Gutiérrez Megías, L. Alfonso Ureña-López, and Eugenio Martínez Cámara. 2024. [Sinai at semeval-2024 task 8: Fine-tuning on words and perplexity as features for detecting machine written text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1516–1521, Mexico City, Mexico. Association for Computational Linguistics.
- Abhimanyu Hans, Avi Schwarzschild, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, and Tom Goldstein. 2024. Spotting llms with binoculars: Zero-shot detection of machine-generated text. *arXiv preprint arXiv:2401.12070*.
- Mohammad Heydari Rad, Farhan Farsi, Shayan Bali, Romina Etezadi, and Mehrnoush Shamsfard. 2024. [Rfbes at semeval-2024 task 8: Investigating syntactic and semantic features for distinguishing ai-generated and human-written texts](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 437–441, Mexico City, Mexico. Association for Computational Linguistics.
- Benjamin D Horne, Jeppe Nørregaard, and Sibel Adali. 2019. Robust fake news detection over time and attack. *ACM Transactions on Intelligent Systems and Technology (TIST)*, 11(1):1–23.
- Daphne Ippolito, Daniel Duckworth, Chris Callison-Burch, and Douglas Eck. 2019. Automatic detection of generated text is easiest when humans are fooled. *arXiv preprint arXiv:1911.00650*.
- Ganesh Jawahar, Muhammad Abdul Mageed, and VS Laks Lakshmanan. 2020. Automatic detection of machine generated text: A critical survey. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 2296–2309.
- Ram Mohan Rao Kadiyala. 2024. [Rkadiyala at semeval-2024 task 8: Black-box word-level text boundary detection in partially machine generated texts](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 498–506, Mexico City, Mexico. Association for Computational Linguistics.
- Ryuto Koike, Masahiro Kaneko, and Naoaki Okazaki. 2023. Outfox: Llm-generated essay detection through in-context learning with adversarially generated examples. *arXiv preprint arXiv:2307.11729*.
- Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. 2023. Paraphrasing evades detectors of ai-generated text, but retrieval is an effective defense. *arXiv preprint arXiv:2303.13408*.
- Tharindu Kumarage, Joshua Garland, Amrita Bhat-tacharjee, Kirill Trapeznikov, Scott Ruston, and Huan Liu. 2023. Stylometric detection of ai-generated text in twitter timelines. *arXiv preprint arXiv:2303.03697*.
- Theodora Kyriakou, Ioannis Maslaris, and Avi Arampatzis. 2024. [Duth at semeval 2024 task 8: Comparing classic machine learning algorithms and llm based methods for multigenerator, multidomain and multilingual machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1069–1075, Mexico City, Mexico. Association for Computational Linguistics.
- Joseph Larson and Francis Tyers. 2024. [Team jelarson at semeval 2024 task 8: Predicting boundary line between human and machine generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 464–471, Mexico City, Mexico. Association for Computational Linguistics.
- Jenny S Li, John V Monaco, Li-Chiou Chen, and Charles C Tappert. 2014. Authorship authentication using short messages from social networking sites. In *2014 IEEE 11th International Conference on e-Business Engineering*, pages 314–319. IEEE.
- Kevin Li, Kenan Hasanaliyev, Sally Zhu, George Alshuler, Alden Eberts, Eric Chen, Kate Wang, Emily Xia, Eli Browne, Ian Chen, and Umut Eren. 2024. [Team mlab at semeval-2024 task 8: Analyzing encoder embeddings for detecting llm-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1474–1478, Mexico City, Mexico. Association for Computational Linguistics.
- Xiaoming Liu, Zhaohan Zhang, Yichen Wang, Hang Pu, Yu Lan, and Chao Shen. 2022. Coco: Coherence-enhanced machine-generated text detection under data limitation with contrastive learning. *arXiv preprint arXiv:2212.10341*.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.
- Anand Kumar M, Abhin B, and Sidhaarth Murali. 2024. [Scalar at semeval-2024 task 8: Unmasking the machine : Exploring the power of roberta ensemble for detecting machine generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1124–1128, Mexico City, Mexico. Association for Computational Linguistics.

- Dominik Macko, Robert Moro, Adaku Uchendu, Jason Lucas, Michiharu Yamashita, Matúš Pikuliak, Ivan Srba, Thai Le, Dongwon Lee, Jakub Simko, and Maria Bielikova. 2023. [MULTITuDE: Large-scale multilingual machine-generated text detection benchmark](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9960–9987, Singapore. Association for Computational Linguistics.
- Dominik Macko, Robert Moro, Adaku Uchendu, Ivan Srba, Jason Samuel Lucas, Michiharu Yamashita, Nafis Irtiza Tripto, Dongwon Lee, Jakub Simko, and Maria Bielikova. 2024. Authorship obfuscation in multilingual machine-generated text detection. *arXiv preprint arXiv:2401.07867*.
- Teodor-George Marchitan, Claudiu Creanga, and Liviu P. Dinu. 2024. [Team unibuc - nlp at semeval-2024 task 8: Transformer and hybrid deep learning based models for machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 390–398, Mexico City, Mexico. Association for Computational Linguistics.
- Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D. Manning, and Chelsea Finn. 2023. [Detectgpt: Zero-shot machine-generated text detection using probability curvature](#). *CoRR*, abs/2301.11305.
- Shaoor Munir, Brishna Batool, Zubair Shafiq, Padmini Srinivasan, and Fareed Zaffar. 2021. Through the looking glass: Learning to attribute synthetic text generated by language models. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 1811–1822.
- ronghao pan, José Antonio García-Díaz, Pedro José Vivancos-Vicente, and Rafael Valencia-García. 2024. [Umuteam at semeval-2024 task 8: Combining transformers and syntax features for machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 683–688, Mexico City, Mexico. Association for Computational Linguistics.
- Kseniia Petukhova, Roman Kazakov, and Ekaterina Kochmar. 2024. [Petkaz at semeval-2024 task 8: Can linguistics capture the specifics of llm-generated text?](#) In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1129–1136, Mexico City, Mexico. Association for Computational Linguistics.
- Valentin Pickard and Hoa Do. 2024. [Tuesents at semeval-2024 task 8: Predicting the shift from human authorship to machine-generated output in a mixed text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 816–819, Mexico City, Mexico. Association for Computational Linguistics.
- Srikar Kashyap Pulipaka, Shrirang Mhalgi, Joseph Larson, and Sandra Kübler. 2024. [Semeval task 8: A comparison of traditional and neural models for detecting machine authored text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1015–1020, Mexico City, Mexico. Association for Computational Linguistics.
- Sadiya Sayara Chowdhury Puspo, Md Nishat Raihan, Dhiman Goswami, Al Nahian Bin Emran, Amrita Ganguly, and Özlem Uzuner. 2024. [Masontigers at semeval-2024 task 8: Performance analysis of transformer-based models on machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1354–1362, Mexico City, Mexico. Association for Computational Linguistics.
- Xiaoyan Qu and Xiangfeng Meng. 2024. [Tm-trek at semeval-2024 task 8: Towards llm-based automatic boundary detection for human-machine mixed text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 696–701, Mexico City, Mexico. Association for Computational Linguistics.
- Areeg Fahad Rasheed and M. Zarkoosh. 2024. [Mashee at semeval-2024 task 8: The impact of samples quality on the performance of in-context learning for machine text classification](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 60–63, Mexico City, Mexico. Association for Computational Linguistics.
- MohammadHossein Rezaei, Yaeun Kwon, Reza Sanayei, Abhyuday Singh, and Steven Bethard. 2024. [Clulab-uofa at semeval-2024 task 8: Detecting machine-generated text using triplet-loss-trained text similarity and text classification](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1509–1515, Mexico City, Mexico. Association for Computational Linguistics.
- Rafael Rivera Soto, Kailin Koch, Aleem Khan, Barry Chen, Marcus Bishop, and Nicholas Andrews. 2024. Few-shot detection of machine-generated text using style representations. *arXiv e-prints*, pages arXiv–2401.
- alberto rodero, Jacinto Mata, and Victoria Pachón Álvarez. 2024. [Boosting ai-generated text detection with multimodal models and optimized ensembles](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 832–839, Mexico City, Mexico. Association for Computational Linguistics.
- Shubhashis Roy Dipta and Sadat Shahriar. 2024. [Hu at semeval-2024 task 8a: Can contrastive learning learn embeddings to detect machine-generated text?](#) In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 472–478, Mexico City, Mexico. Association for Computational Linguistics.

- Areg Mikael Sarvazyan, José Ángel González, and Marc Franco-Salvador. 2024. [Genaios at semeval-2024 task 8: Detecting machine-generated text by mixing language model probabilistic features](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 101–107, Mexico City, Mexico. Association for Computational Linguistics.
- Surbhi Sharma and Irfan Mansuri. 2024. [Team innovative at semeval-2024 task 8: Multigenerator, multidomain, and multilingual black-box machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1161–1165, Mexico City, Mexico. Association for Computational Linguistics.
- Zhouxing Shi, Yihan Wang, Fan Yin, Xiangning Chen, Kai-Wei Chang, and Cho-Jui Hsieh. 2023. Red teaming language model detectors with language models. *arXiv preprint arXiv:2305.19713*.
- Alexander Shirnin, Nikita Andreev, Vladislav Mikhailov, and Ekaterina Artemova. 2024. [Aipom at semeval-2024 task 8: Detecting ai-produced outputs in m4](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1678–1683, Mexico City, Mexico. Association for Computational Linguistics.
- Lei Shu, Liangchen Luo, Jayakumar Hoskore, Yun Zhu, Canoe Liu, Simon Tong, Jindong Chen, and Lei Meng. 2023. RewritelM: An instruction-tuned large language model for text rewriting. *arXiv preprint arXiv:2305.15685*.
- Marco Siino. 2024. [Badrock at semeval-2024 task 8: Distilbert to detect multigenerator, multidomain and multilingual black-box machine-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 239–245, Mexico City, Mexico. Association for Computational Linguistics.
- Irene Solaiman, Miles Brundage, Jack Clark, Amanda Askeel, Ariel Herbert-Voss, Jeff Wu, Alec Radford, Gretchen Krueger, Jong Wook Kim, Sarah Kreps, et al. 2019. Release strategies and the social impacts of language models. *arXiv preprint arXiv:1908.09203*.
- Michal Spiegel and Dominik Macko. 2024. [Kinit at semeval-2024 task 8: Fine-tuned llms for multilingual machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 545–551, Mexico City, Mexico. Association for Computational Linguistics.
- Harald Stiff and Fredrik Johansson. 2022. Detecting computer-generated disinformation. *International Journal of Data Science and Analytics*, 13(4):363–383.
- Daniel Stuhlinger and Aron Winkler. 2024. [TueciCl at semeval-2024 task 8: Resource-efficient approaches for machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1608–1612, Mexico City, Mexico. Association for Computational Linguistics.
- Jinyan Su, Terry Yue Zhuo, Di Wang, and Preslav Nakov. 2023. Detectllm: Leveraging log rank information for zero-shot detection of machine-generated text. *arXiv preprint arXiv:2306.05540*.
- Bao Tran and Nhi Tran. 2024. [Newbieml at semeval-2024 task 8: Ensemble approach for multidomain machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 347–353, Mexico City, Mexico. Association for Computational Linguistics.
- Hanh Thi Hong Tran, Tien Nam Nguyen, Antoine Doucet, and Senja Pollak. 2024. [L3i++ at semeval-2024 task 8: Can fine-tuned large language model detect multigenerator, multidomain, and multilingual black-box machine-generated text?](#) In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 13–21, Mexico City, Mexico. Association for Computational Linguistics.
- Adaku Uchendu, Thai Le, and Dongwon Lee. 2023. Attribution and obfuscation of neural text authorship: A data mining perspective. *ACM SIGKDD Explorations Newsletter*, 25(1):1–18.
- Adaku Uchendu, Thai Le, Kai Shu, and Dongwon Lee. 2020. Authorship attribution for neural text generation. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pages 8384–8395.
- Adaku Uchendu, Zeyu Ma, Thai Le, Rui Zhang, and Dongwon Lee. 2021. Turingbench: A benchmark environment for turing test in the age of neural text generation. *arXiv preprint arXiv:2109.13296*.
- Ashok Urlana, Aditya Saibewar, Bala Mallikarjunarao Garlapati, Charaka Vinayak Kumar, Ajeet Singh, and Srinivasa Rao Chalamala. 2024. [Trustai at semeval-2024 task 8: A comprehensive analysis of multi-domain machine generated text detection techniques](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 914–921, Mexico City, Mexico. Association for Computational Linguistics.
- Andric Valdez, Fernando Márquez, Jorge Pantaleón, Helena Gómez, and Gemma Bel-Enguix. 2024. [iimasnlp at semeval-2024 task 8: Unveiling structure-aware language models for automatic generated text identification](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1099–1103, Mexico City, Mexico. Association for Computational Linguistics.

- Saranya Venkatraman, Adaku Uchendu, and Dongwon Lee. 2023. Gpt-who: An information density-based machine-generated text detector. *arXiv preprint arXiv:2310.06202*.
- Anastasia Voznyuk and Vasily Konovalov. 2024. Deep-pavlov at semeval-2024 task 8: Leveraging transfer learning for detecting boundaries of machine-generated texts. In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1833–1841, Mexico City, Mexico. Association for Computational Linguistics.
- Boxin Wang, Chejian Xu, Shuohang Wang, Zhe Gan, Yu Cheng, Jianfeng Gao, Ahmed Hassan Awadallah, and Bo Li. 2021. Adversarial glue: A multi-task benchmark for robustness evaluation of language models. *arXiv preprint arXiv:2111.02840*.
- Yuxia Wang, Jonibek Mansurov, Petar Ivanov, Jinyan Su, Artem Shelmanov, Akim Tsvigun, Osama Mohammed Afzal, Tarek Mahmoud, Giovanni Puccetti, Thomas Arnold, et al. 2024a. M4gt-bench: Evaluation benchmark for black-box machine-generated text detection. *arXiv preprint arXiv:2402.11175*.
- Yuxia Wang, Jonibek Mansurov, Petar Ivanov, Jinyan Su, Artem Shelmanov, Akim Tsvigun, Chenxi Whitehouse, Osama Mohammed Afzal, Tarek Mahmoud, Toru Sasaki, Thomas Arnold, Alham Aji, Nizar Habash, Iryna Gurevych, and Preslav Nakov. 2024b. M4: Multi-generator, multi-domain, and multilingual black-box machine-generated text detection. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1369–1407, St. Julian’s, Malta. Association for Computational Linguistics.
- Yuchen Wei. 2024. Team at at semeval-2024 task 8: Machine-generated text detection with semantic embeddings. In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 479–483, Mexico City, Mexico. Association for Computational Linguistics.
- Youlin Wu, Kaichun Wang, Kai Ma, Liang Yang, and Hongfei LIN. 2024. Werkzeug at semeval-2024 task 8: Llm-generated text detection via gated mixture-of-experts fine-tuning. In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 534–539, Mexico City, Mexico. Association for Computational Linguistics.
- Zhuohan Xie, Trevor Cohn, and Jey Han Lau. 2023. The next chapter: A study of large language models in storytelling. In *Proceedings of the 16th International Natural Language Generation Conference*, pages 323–351.
- Feng Xiong, Thanet Markchom, Ziwei Zheng, Subin Jung, Varun Ojha, and Huizhi Liang. 2024. Ncl-uor at semeval-2024 task 8: Fine-tuning large language models for multigenerator, multidomain, and multilingual machine-generated text detection. In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 163–169, Mexico City, Mexico. Association for Computational Linguistics.
- Xiaoman Xu, Xiangrun Li, Taihang Wang, Jianxiang Tian, and Ye Jiang. 2024. Team qust at semeval-2024 task 8: A comprehensive study of monolingual and multilingual approaches for detecting ai-generated text. In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 450–457, Mexico City, Mexico. Association for Computational Linguistics.
- Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, and Yejin Choi. 2019. Defending against neural fake news. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pages 9051–9062.
- Wanjun Zhong, Duyu Tang, Zenan Xu, Ruize Wang, Nan Duan, Ming Zhou, Jiahai Wang, and Jian Yin. 2020. Neural deepfake detection with factual structure of text. *arXiv preprint arXiv:2010.07475*.

Appendix

A Method Summary

A.1 Monolingual Human vs, Machine

Team Genaios_{STA_mono:1} (Sarvazyan et al., 2024) achieves the highest accuracy on Subtask A – Monolingual by extracting token-level probabilistic features using four Llama-2 models: Llama-2-7b, Llama-2-7b-chat, Llama-2-13b, and Llama-2-13b-chat. For each token they compute the log probability of the observed token, the log probability of the token predicted by each of the language models, and the entropy of the distribution. These features are then fed to a Transformer Encoder trained in a supervised fashion to detect synthetic text.

Team USTC-BUPT_{STA_mono:2} (Guo et al., 2024) incorporates domain adversarial neural networks into the task of machine-generated text detection to reach the second position in the ranking of Subtask A – Monolingual. They add a gradient reversal layer on top of the baseline, a supervised classifier based on RoBERTa. In addition, they exploit domain labels to enhance the transferability of learning between training and testing datasets. Their architecture is based on RoBERTa and adds two classification heads, one for category classification (human or synthetic) and one for domain classification (e.g. news, essays, etc.), the former uses an MLP layer and the latter is composed of an MLP together with a gradient reversal layer. Finally, the loss is also adapted by summing together the category and the domain losses. The submission evaluated an improvement of approximately 8% compared to the baseline.

Team PetKaz_{STA_mono:12} (Petukhova et al., 2024) uses a PLM, RoBERTa-base, fine-tuned for synthetic text detection and enhances it with linguistic features, to train a feed-forward binary classifier (human or synthetic). Their final model uses diverse features and notably, they undersample the human data.

Team HU_{STA_mono:17} (Roy Dipta and Shahriar, 2024) Adopts an architecture trained with a contrastive learning approach based on fine-tuning *sentence-transformers/all-mpnet-base-v2*. The model is trained on an augmented dataset obtained by paraphrasing sentences in the training set.

Team TrustAI_{STA_mono:20} (Urlana et al., 2024) tries two approaches: (a) an ensemble approach with the combination of Multinomial Naive Bayes, LGBM Classifier (lightGBM classifier) and SGD classifier. Each is trained on the concatenation of tf-idf and spaCy embeddings obtained from the Subtask A – Monolingual dataset and (b) a synthetic text classifier based on RoBERTa fine-tuned first with the outputs of the 1.5B-parameter GPT-2 model and subsequently on the Subtask A – monolingual dataset. They show that exploring methodologies with different assumptions helps identify the best performing approach.

Team RFBES_{STA_mono:23} (Heydari Rad et al., 2024) Both semantic and syntactic considerations were taken into account. For semantic analysis, emphasis was placed on smaller text segments rather than the entire document, operating under the belief that AI models could produce similarly coherent long texts as humans. To achieve this, the XLM-RoBERTa model was employed. Regarding syntactic analysis, a stacked bidirectional LSTM model was used to categorize texts based on their grammatical patterns using UPOS tags. Interestingly, no significant differences in UPOS tag distribution between AI-generated and human-written texts were revealed by the findings.

Team L3i++_{STA_mono:24} (Tran et al., 2024) Proposes a comparative study among 3 groups of methods to detect synthetic texts: 5 likelihood-based methods; 2 fine-tuned sequence-labeling language models (RoBERTa, XLM-RoBERTa); and a fine-tuned large language model, llama-2-7b. LLaMA 2 outperforms the rest and accurately detects machine-generated texts.

Team art-nat-HH_{STA_mono:25} (Ciccarelli et al., 2024) fine-tunes a RoBERTa model pre-trained for AI-detection and combines it with a set of linguistic features: syntactic, lexical, probabilistic and stylistic. To improve the classifier, they train two separate neural networks on these features, one for each class predicted by the RoBERTa-based classifier.

Team Unibuc - NLP_{STA_mono:28} (Marchitan et al., 2024) fine-tunes a Transformer-based model with a MLP as a classification head. They combine the datasets of Subtask A – monolingual and Subtask B to obtain a larger training set.

NewbieML_{STA_mono:30} (Tran and Tran, 2024) embeds texts with Longformer-large. Then they Ensemble SVM, LogisticRegression and XGBoost with, as a meta model, a KNN.

Team QUST_{STA_mono:31} (Xu et al., 2024) experiments with multiple models on a dataset extended through data augmentation. They select the two best-performing models for ensembling: (1) a fine-tuned RoBERTa model, combined with the Multiscale Positive-Unlabeled (MPU) training and (2) a DeBERTa model. They use these two for model fusion through stacking ensemble.

Team NootNoot_{STA_mono:39} (Bahad et al., 2024) carefully fine-tunes a RoBERTa-base model to classify human written and synthetic texts.

Team Mast Kalandar_{STA_mono:40} (Bafna et al., 2024) trains a classifier that uses a frozen RoBERTa model with an LSTM head to classify human vs machine written texts.

Team I2C-Huelva_{STA_mono:41} (rodero et al., 2024) proposes a method to use multimodal models together with text analysis to enhance synthetic text detection. To mix the two approaches they explore ensemble by testing several voting methods.

Team Werkzeug_{STA_mono:45} (Wu et al., 2024) uses Roberta-large and XLM-roberta-large to encode texts. To address the anisotropic embedding space created by transformer-based language models, they employ several learnable parametric whitening (PW) transformation. They show that addressing the anisotropy of the embedding space improves accuracy in detecting synthetic text.

Team NCL-UoR_{STA_mono:50} (Xiong et al., 2024) fine-tunes several PLMs including XLM-RoBERTa, RoBERTa with Low-Rank Adaptation (LoRA) and DistilBERT. Finally, they use majority voting ensembling with XLM-RoBERTa and LoRA-RoBERTa. To confirm that ensembling is a strong technique to boost synthetic text classification accuracy.

Team Sharif-MGTD_{STA_mono:51} (Ebrahimi et al., 2024) carefully fine-tunes RoBERTa-base for synthetic text detection, show that pre-trained language models are a versatile approach.

Team BadRock_{STA_mono:*} (Siino, 2024) is based on a fine-tuning of a DistilBERT trained on the SST-2 dataset.

Team Collectivized Semantics_{STA_mono:62} (Datta et al., 2024) fine-tunes Roberta-base using Ada-LoRa and uses the weighted sum of all the layer hidden states' mean as features to train a classifier. They show that exploiting the knowledge at all layers of encoder language models helps when detecting synthetic texts.

Team IUCL_{STA_mono:63} (Pulipaka et al., 2024) tries both classical ML classifiers, Naive Bayes and Decision Trees as well as fine-tuning transformers and they conclude that fine-tuned RoBERTa is best among the methods they try.

Team SINAI_{STA_mono:67} (Gutiérrez Megías et al., 2024) compares three methods: (a) supervised classification, based on fine-tuning the XLM-RoBERTa-Large language model; (b) likelihood-based methods, using GPT-2 to compute the perplexity of each text and use this perplexity as a score; (c) a hybrid approach that merges text with its perplexity value into a classification head. The choice of a mixed approach proves effective in improving synthetic text detection accuracy.

Team MasonTigers_{STA_mono:71} (Puspo et al., 2024) experiments with different transformer-based models: Roberta, DistilBERT, ELECTRA and ensembles these models. They also experiment with zero-shot prompting and finetuning FlanT5. Further confirming that ensembling is a strong methodology for detecting synthetic texts.

Team AT_{STA_mono:72} (Wei, 2024) adopts three different semantic embedding algorithms, GLOVE, n-gram embeddings and SentenceBERT as well as their concatenation. The author finds that these pre-trained embeddings, while fast to compute, are not as effective as a fine-tuned RoBERTa model.

Team DUTh_{STA_mono:73} (Kyriakou et al., 2024) experiments with several supervised classification models based on PLMs. Finally, they opt for a fine-tuned mBERT trained for 5 epochs. This approach shows how PLMs fine-tuning is a versatile approach that can be effective when detecting synthetic texts.

Team surbhi_{STA_mono:74} (Sharma and Mansuri, 2024) creates two sets of features (a) stylometric features based on the length of text, the number of words, the average length of words, the number of short words, the proportion of digits and capital letters, individual letters and digits frequencies, hapax-legomena, a measure of text richness, and the frequency of 12 punctuation marks and (b) n-grams: frequencies of the 100 most frequent character-level bi-grams and tri-grams; (c) the output probabilities of fine-tuned Roberta model. Each set of features is used to train a classifier and finally, stylometric and n-gram features are chosen as the best-performing ones. They prove that more classical features can still be valuable when attempting the detection of synthetic text.

Team Kathlalu_{STA_mono:76} (Cao et al., 2024) investigates two methods for constructing a binary classifier to distinguish between human-generated and machine-generated text. The main emphasis is on a straightforward approach based on Zipf's law, which, despite its simplicity, achieves a moderate level of performance. Additionally, they briefly discuss experimentation with the utilization of unigram word counts.

Team KInIT_{STA_mono:77} (Spiegel and Macko, 2024) uses two approaches: (a) an ensemble using two-step majority voting for predictions, consisting of 2 LLMs (Falcon-7B and Mistral-7B) fine-tuned using the train set only; (b) 3 zero-shot statistical methods (Entropy, Rank, Binoculars) using Falcon-7B and Falcon-7B-Instruct for calculating the metrics. For classification they use per-language threshold calibration, showing that likelihood-based methods are a viable solution to detect machine-written texts.

Team iimasNLP_{STA_mono:78} (Valdez et al., 2024) fine-tune 4 different language models to identify human and machine generated text, ERNIE, SpanBERT, ConvBERT and XLNet. They find out that RoBERTa is a stronger classifier. In general this shows how fine-tuning PLMs is an effective approach to identify synthetic text.

Team Groningen-F_{STA_mono:86} (Donker et al., 2024) leverage features including tense of the sentence, the voice of the sentence, the sentiment of the sentence, and the number of pronouns vs. proper nouns on the basis of SVM and FFNN models. The hypothesis here is that traditional models may generalize better than LLMs. It is more computationally effective than LLMs.

Team RUG-D_{STA_mono:100} (Brekhof et al., 2024) fine-tunes different DeBERTa models on a dataset extended with additional synthetic samples. Showing that PLMs fine-tuning is a versatile approach that can be effective in the detection of synthetic texts.

Team RUG-5_{STA_mono:101} (Darwinkel et al., 2024) fine-tunes different pre-trained models for synthetic text classification, distilbert-base-cased for the monolingual tasks and distilbert-base-multilingual-cased for the multilingual ones. Moreover, they explore the use of a Random Forest classifier using frozen distilbert-base-cased embeddings concatenated with 20 linguistic and stylistic features. This approach shows how choosing the right PLMs is crucial for better performance in a given task.

Team Mashee_{STA_mono:115} (Rasheed and Zarkoosh, 2024) selects high-quality and low-quality samples using a Chi-square test and adopts the selected samples for few-shot classification using the FlanT5-Large language model. This approach shows how few-shot methodologies can benefit from a careful example selection.

Team TueCICL_{STA_mono:116} (Stuhlinger and Winkler, 2024) uses a Character-level LSTM with pre-trained word2vec embeddings as input to train synthetic text detector. Doing so, they show how one does not necessarily have to use transformers.

Team RUG-1_{STA_mono:117} (Alecakir et al., 2024) combines a linear model with document-level features and token-level features that are first passed through an LSTM. Through this methodology, they leverage both local (token-level) and global (document-level) information to identify human-written and synthetic texts.

Team CUNLP_{STA_mono:unknown}¹ (Aggarwal and Sachdeva, 2024) involved employing a range of machine learning techniques, including logistic regression, transformer models, attention mechanisms, and unsupervised learning methods. Through rigorous experimentation, they identified key features influencing classification accuracy, namely text length, vocabulary richness, and coherence. Notably, the highest classification accuracy was achieved by integrating transformer models with TF-IDF representation and feature engineering. However, it is essential to note that this approach demanded substantial computational resources due to the complexity of transformer models and the incorporation of TF-IDF. Additionally, their investigation encompassed a thorough exploration of various ML algorithms, extensive hyperparameter tuning, and optimization techniques. Furthermore, they conducted detailed exploratory data analysis to gain insights into the structural and lexical characteristics of the text data.

A.2 Multilingual Human vs Machine

Team USTC-BUPT_{STA_Multi:1} (Guo et al., 2024) secured the top position. They initially detect the language of the input text. For English text, they average embeddings from Llama-2-70B, followed by classification through a two-stage CNN. For texts in other languages, the classification problem is transformed into fine-tuning a next-token prediction task using the mT5 model, incorporating special tokens for classification. Their approach integrates both monolingual and multilingual strategies, exploiting large language models for direct embedding extraction and model fine-tuning. This enables the system to adeptly handle text classification across a diverse range of languages, especially those with fewer resources.

Team FI Group_{STA_Multi:2} (Ben-Fares et al., 2024) came in second place. Their methodology began with analyzing latent space distinctions between human and AI-generated texts using Sentence-BERT, hypothesizing that syntax and writing style differences are key. They utilized a hierarchical fusion strategy to adaptively fuse representations from different BERT layers, focusing on syntax over semantics. By classifying each token as Human or AI, their model captures detailed text structures, leveraging the XLM-RoBERTa-Large model for robust multilingual performance.

Team KInIT_{STA_Multi:3} (Spiegel and Macko, 2024) placed third by employing an ensemble of two fine-tuned LLMs (Falcon-7B and Mistral-7B) and three zero-shot statistical methods, using a two-step majority voting system. This unique combination of fine-tuned and statistical methods, complemented by language identification and per-language threshold calibration, showcases their innovative approach to integrating diverse techniques for enhanced classification accuracy.

Team L3i++_{STA_Multi:5} (Tran et al., 2024) explored a comparative study among metric-based models, fine-tuned sequence-labeling language models, and a large-scale LLM, finding LLaMA-2 to outperform others in detecting machine-generated texts. Their methodological diversity and comprehensive analysis underline the strengths of fine-tuning LLMs for complex classification tasks across languages.

Team QUST_{STA_Multi:6} (Xu et al., 2024) employed a fine-tuned XLM-RoBERTa model within a stacking ensemble framework, incorporating the MPU framework and DeBERTa model. Their approach emphasizes the efficacy of model fusion and fine-tuning on a multilingual dataset, highlighting the potential of ensemble strategies in enhancing model performance.

Team AIpom_{STA_Multi:9} (Shirnin et al., 2024) utilized a LoRA-Finetuned LLM for classifying texts as real or fake, achieving notable results with a limited dataset. Their unique approach of using an LLM as a classifier, despite an accidental label swap during training, emphasizes the versatility and potential of LLMs in unconventional scenarios.

Team RFBES_{STA_Multi:10} (Heydari Rad et al., 2024) Both semantic and syntactic considerations were taken into account. For semantic analysis, emphasis was placed on smaller text segments rather than the

¹Team CUNLP submitted results for development set, but no submissions for the test set, resulting unknown valid rank.

entire document, operating under the belief that AI models could produce similarly coherent long texts as humans. To achieve this, the XLM-RoBERTa model was employed. Regarding syntactic analysis, a stacked bidirectional LSTM model was used to categorize texts based on their grammatical patterns using UPOS tags. Interestingly, no significant differences in UPOS tag distribution between AI-generated and human-written texts were revealed by the findings.

Team SINAI_{STA_Multi:21} (Gutiérrez Megías et al., 2024) compared various systems before settling on a fusion model that integrates text with perplexity values for classification. Their comprehensive approach, blending fine-tuning with innovative use of perplexity, offers insightful perspectives on leveraging multiple data dimensions for classification.

Team Unibuc-NLP_{STA_Multi:22} (Marchitan et al., 2024) focused on exploring different methods of layer selection and fine-tuning within a transformer-based architecture. Their pursuit of optimizing layer interactions for classification tasks highlights the importance of fine-tuning strategies in achieving model effectiveness.

Team Werkzeug_{STA_Multi:30} (Wu et al., 2024) applied parametric whitening transformations under a mixture-of-experts architecture to address text embedding anisotropy issues. Their methodological innovation, aimed at capturing a broader range of language styles, underscores the potential of advanced architectures in improving classification accuracy.

Team RUG-5_{STA_Multi:32} (Darwinkel et al., 2024) augmented DistilBERT with an additional layer for classification, exploring linguistic-stylistic features alongside Random Forest classifiers. Their approach of blending traditional ML techniques with PLMs offers a novel perspective on enhancing text classification through feature integration.

Team DUTH_{STA_Multi:33} (Kyriakou et al., 2024) compared machine learning algorithms and LLMs, ultimately selecting a fine-tuned XLM-RoBERTa model. Their comparative analysis provides valuable insights into the effectiveness of different methodologies for text classification tasks.

Team RUG-D_{STA_Multi:39} (Brekhof et al., 2024) used an ensemble of monolingual and multilingual models, testing the performance impact of additional training data. Their ensemble approach and data augmentation strategy highlight the importance of model and data selection in optimizing classification performance.

Team MasonTigers_{STA_Multi:49} (Puspo et al., 2024) experimented with different transformer models and finetuning strategies, showcasing the effectiveness of ensembling and fine-tuning in addressing classification challenges.

Team TrustAI_{STA_Multi:55} (Urlana et al., 2024) focused on fine-tuning the bert-base-multilingual-cased model, demonstrating the potential of pre-trained models in multilingual text classification tasks.

A.3 Multi-way Detection

Team AISPACE_{STB:1} (Gu and Meng, 2024) achieves the highest performance in this subtask by fine-tuning various encoder and encoder-decoder models, including RoBERTa, DeBERTa, XLNet, Longformer, and T5. They augment the data with instances from Subtask A and explore the effects of different loss functions and learning rate values. Based on this analysis, they leverage a weighted Cross-Entropy loss to balance samples in different classes. Furthermore, they use an ensemble of different fine-tuned models to improve the robustness of the system. The weights of the models in the ensemble are assigned based on their performance on each class rather than their performance on the whole accuracy.

Team Unibuc - NLP_{STB:2} (Marchitan et al., 2024) use a Transformer-based model with a peculiar two-layer feed-forward network as a classification head. They also augment the data with instances from Subtask A monolingual dataset.

Team USTC-BUPT_{STB:3} (Guo et al., 2024) first leverage the ‘Llama-2-70B’ model to obtain embeddings of the tokens in the text and then average them across all tokens. Next, they employ a three-stage classification approach using the CNN classifier.

Firstly, they distinguish between human-generated and machine-generated text using the Llama-2-70B model. Secondly, they categorize ChatGPT and Cohere as a single class for a four-class classification, differentiating them from Davinci, Bloomz, and Dolly. Finally, they perform a binary classification

between ChatGPT and Cohere. Despite solid performance, their method does not require fine-tuning.

Team L3i++_{STB:6} (Tran et al., 2024) conduct a comparative study among three groups of methods: metric-based models, fine-tuned classification language models (RoBERTa, XLM-R), and a fine-tuned LLM, LLaMA-2-7b. They find LLaMA-2 outperforming the methods from the other groups in MGT detection. The team reveals the analysis of errors and various factors in their paper.

Team MLab_{STB:7} (Li et al., 2024) fine-tune DeBERTa and analyze the embeddings from the last layer. They provide insights into the embedding space of the model.

Team Werkzeug_{STB:8} (Wu et al., 2024) utilizes RoBERTa-large and XLM-RoBERTa-large to encode the text. They tackle the problem of anisotropy in text embeddings produced by pre-trained language models (PLMs) by introducing a learnable parametric whitening (PW) transformation. Furthermore, to capture the features of LLM-generated text from different perspectives, they use multiple PW transformation layers as experts under the mixture-of-experts (MoE) architecture equipped with a gating router in their final solution.

Team TrustAI_{STB:14} (Urlana et al., 2024) explore different pretrained and statistical models for detecting synthetic text, ultimately selecting the RoBERTa-base OpenAI Detector for its effectiveness. This model, originally fine-tuned with outputs from the 1.5B-parameter GPT-2 model, is further fine-tuned on the Subtask-B dataset.

Team MGTD4ADL_{STB:17} (Chen et al., 2024) combine traditional Transformer models (RoBERTa-base, RoBERTa-large, GPT-2-small, XLNet, T5-small) with Sentence Transformers(all-mpnet-base-v2 and all-roberta-large-v1). They further diversify their approach by leveraging different data augmentation techniques and experimenting with various loss functions such as Cross-Entropy (CE), Supervised Contrastive Learning (SCL), and Dual Contrastive Loss (DUALCL).

Team scalar_{STB:18} (M et al., 2024) employ an ensemble of three RoBERTa-base models using an individual validation set for each model.

Team UMUTeam₂₃ (pan et al., 2024) use fine-tuned RoBERTa model combined with syntactic features of the text such as word length, part of speech, function word frequency, stop-word ratio, and sentence length.

Team QUST_{STB:36} (Xu et al., 2024) use fine-tuned RoBERTa and DeBERTa models, integrating them through a stacking ensemble technique.

Team MasonTigers_{STB:38} (Puspo et al., 2024) implement an ensemble of 3 PLMs: RoBERTa, DeBERTa, and ELECTRA. Additionally, they employ zero-shot prompting and use a fine-tuned FLAN-T5 model.

Team RUG-5_{STB:41} (Darwinkel et al., 2024) expands the architecture of DistilBERT models by adding an additional classification layer that incorporates 20 linguistic-stylistic features. They also explore the use of Random Forest classifier on top of embeddings from DistilBERT combined with the same set of linguistic-stylistic features.

Team RUG-D_{STB:44} (Brekhof et al., 2024) focus on fine-tuning DeBERTa models.

Team Groningen-F_{STB:45} (Donker et al., 2024) trained traditional machine learning models (SVM and FFNN) with features including tense of the sentence, the voice of the sentence, the sentiment of the sentence, and the number of pronouns vs. proper nouns.

Team DUTh_{STB:49} (Kyriakou et al., 2024) explore traditional machine learning algorithms along with BERT for their task. Ultimately, they proceed with BERT fine-tuned for 5 epochs.

Team AT_{STB:58} (Wei, 2024) adopts three different semantic embedding algorithms, GLOVE, n-gram embeddings and SentenceBERT as well as their concatenation to identify the generator in Subtask B. The author finds that these pre-trained embeddings, while fast to compute, are not as effective as a fine-tuned RoBERTa model.

Team iimasNLP_{STB:61} (Valdez et al., 2024) fine-tune 4 different language models to classify text generated by different models: ERNIE, SpanBERT, ConvBERT and XLNet. They find out that RoBERTa is a stronger classifier. In general this shows how fine-tuning PLMs is an effective approach to identify the generator model.

Team CLULab-UofA_{STB:62} (Rezaei et al., 2024) combine LLM fine-tuning with contrastive learning, specifically using triplet loss.

A.4 Boundary Identification

Team TM-TREK_{STC:1} (Qu and Meng, 2024) achieved the highest performance in Subtask C by employing an ensemble of models including Longformer, Bigbird, and XLNet for long-text sequence labeling. A simple voting mechanism was used to aggregate the output logits. Their innovative strategy also involved integrating LSTM and CRF layers atop various pre-trained language models (PLMs), along with continuous pretraining, fine-tuning, and utilizing dice loss functions to enhance model performance.

Team AIpom_{STC:2} (Shirnin et al., 2024) introduced a two-stage pipeline that combines outputs from an instruction-tuned, decoder-only model (Mistral-7B-OpenOrca) with two encoder-only sequence taggers. Initially, they trained an instruction-tuned autoregressive model to insert a [BREAK] token into input texts, delineating human-written parts from machine-generated ones. Subsequently, these annotated texts were processed by an encoder-based model for sequence tagging, differentiating human-written tokens (0) from machine-generated tokens (1). An additional encoder trained on a blend of raw and annotated texts further refined sequence tagging. The average change point positions predicted by both encoders served as the final boundary estimation.

Team USTC-BUPT_{STC:3} (Guo et al., 2024) approached the task as a token classification challenge, opting to fine-tune a DeBERTa model enhanced by data augmentation techniques derived from the training set. They reported that DeBERTa-base outperformed other models, and explored the efficacy of sequence labeling (e.g., BIOS) in detecting boundaries within mixed texts. The potential of various layers, including CRF and Dropout, was also examined for their impact on system performance.

Team RKadiyala_{STC:6} (Kadiyala, 2024) focused on fine-tuning various encoder-based models appended with a Conditional Random Field (CRF) layer, noting that DeBERTa-V3 yielded the best results on the development set.

Team DeepPavlov_{STC:7} (Voznyuk and Konovalov, 2024) fine-tuned the DeBERTa-v3 model using a specially prepared dataset with augmented texts, created by modifying prefixes and suffixes of original texts. They emphasized the importance of augmented data quality in achieving a mean absolute error (MAE) of 15.20903.

Team RUG-5_{STC:17} (Darwinkel et al., 2024) utilized an augmented Longformer model, incorporating extra features into the output state of each token to enrich them with contextual information. This approach aimed at improving token-level classification by leveraging linguistic-stylistic features beyond simple PLM optimization.

Team TueCICL_{STC:22} (Stuhlinger and Winkler, 2024) experimented with character-level LSTMs and LSTMs using pretrained Word2Vec embeddings, demonstrating that smaller models could compete with transformer models in the boundary detection task.

Team jelarson_{STC:25} (Larson and Tyers, 2024) explored rule-based methods and linear regression techniques, identifying specific patterns in the training data that could inform better data collection practices, such as ensuring a more randomized and unbiased dataset.

Team TueSents_{STC:26} (Pickard and Do, 2024) extracted textual features at the sentence level using tools like SpaCy and trained a lightweight BiLSTM model for boundary prediction, achieving an accuracy of 0.7 and MAE of less than 0.5 on the development set.

Team MasonTigers_{STC:27} (Puspo et al., 2024) combined TF-IDF, PPMI, and RoBERTa features with linear regression and Elastic Net, culminating in an ensemble approach based on a weighted development set.

Team Unibuc-NLP_{STC:28} (Marchitan et al., 2024) framed the task as a token classification problem, merging character-level features (extracted via CNN) and word embeddings within a BiLSTM model, further exploring the addition of CRF for enhanced performance.